

One-Way Sensitivity Analysis Report

Biomarker-Guided Strategies for Selecting Immunotherapy Candidates in Metastatic MSS/pMMR Colorectal Cancer

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1 Overview

This one-way sensitivity analysis (OWSA) evaluates the robustness of the cost-effectiveness results for three biomarker-guided strategies (CRP, TLR, and TMB/BRAF) compared to standard of care for selecting metastatic microsatellite stable (MSS)/proficient mismatch repair (pMMR) colorectal cancer patients for immunotherapy treatment.

The analysis examines how variations in individual model parameters affect the net monetary benefit (NMB) and optimal strategy selection, providing insights into which parameters have the greatest impact on decision uncertainty.

This report contains:

- **Technical Details:** Analysis specifications and methodology
- **Parameter Overview:** Description of the key parameters varied in the sensitivity analysis
- **Tornado Diagrams:** Visual representation of parameter impact for each strategy
- **Line Plot Analysis:** Parameter sensitivity across the full range for NMB, costs, and effectiveness
- **Robustness Analysis:** Assessment of strategy switching and decision uncertainty

2 Technical Details

2.1 Analysis Specifications

Model Configuration:

- Total model evaluations: 124
- Strategies evaluated: 4 (control, crp, tlr, tmb_braf)
- Parameters tested: 14
- Time horizon: 520 weeks (10 years)
- Discount rate: 4% per annum
- Willingness-to-pay threshold: €51,000
- Parameter variation range: $\pm 20\%$

Parameter Constraints:

- Utility parameters capped at maximum value of 1.0
- All cost parameters constrained to non-negative values

Methodology:

- One-way sensitivity analysis varies each parameter individually
- All other parameters held at base case values during each test
- Net Monetary Benefit calculated as: $NMB = \text{Effect} \times WTP - \text{Cost}$
- Tornado diagrams ordered by magnitude of parameter impact
- Line plots show outcome response across actual parameter values
- Three outcome measures: Net Monetary Benefit, Total Costs, and Total Effectiveness

2.2 Computational Environment

Analysis completed on: December 22, 2025 **R version:** R version 4.5.1 (2025-06-13 ucrt)

Repository: ben-geisler/METIMMOX-1 **Report version:** 2.0

3 Analysis Parameters

Table 1: Parameters Included in One-Way Sensitivity Analysis

Group	Parameter Description	Base	Min (-20%)	Max (+20%)
Drug costs	Nivolumab cost per administration	13923	11138	16708
	FLOX chemotherapy cost per administration	427	342	512
Test costs	Next-generation sequencing (TMB/BRAF testing)	1439	1151	1727
	Computed tomography imaging	386	309	463
	Blood test cost	0.00	0.00	0.00
Other costs	Standard outpatient visit	33.00	26.40	39.60
	Comprehensive baseline visit	530	424	636
	Follow-up visit cost	33.00	26.40	39.60
	End-of-life care costs	13803	11042	16564
Utilities	Utility - progression-free state	0.91	0.73	1.00
	Utility - progressed disease state	0.90	0.72	1.00
Prevalence	CRP biomarker positive prevalence	0.34	0.27	0.41
	TLR biomarker positive prevalence	0.65	0.52	0.78
	TMB/BRAF biomarker positive prevalence	0.45	0.36	0.54
Survival model	Parametric survival distribution (OS & PFS)	gamma	See section 7	See section 7

4 Tornado Diagrams

4.1 Standard of Care Strategy

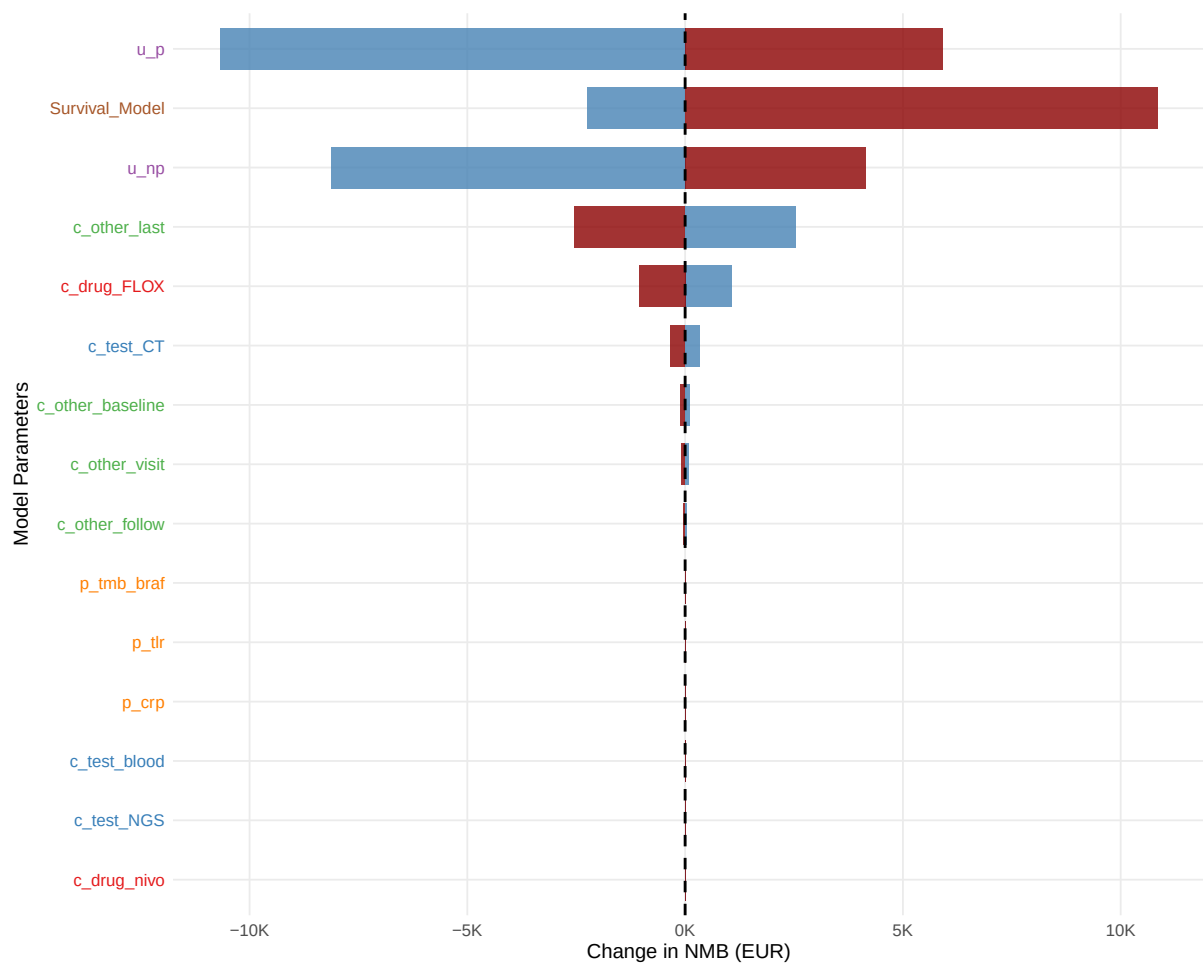


Figure 1: Tornado Diagram - Standard of Care One-Way Sensitivity Analysis

4.2 CRP-guided Strategy

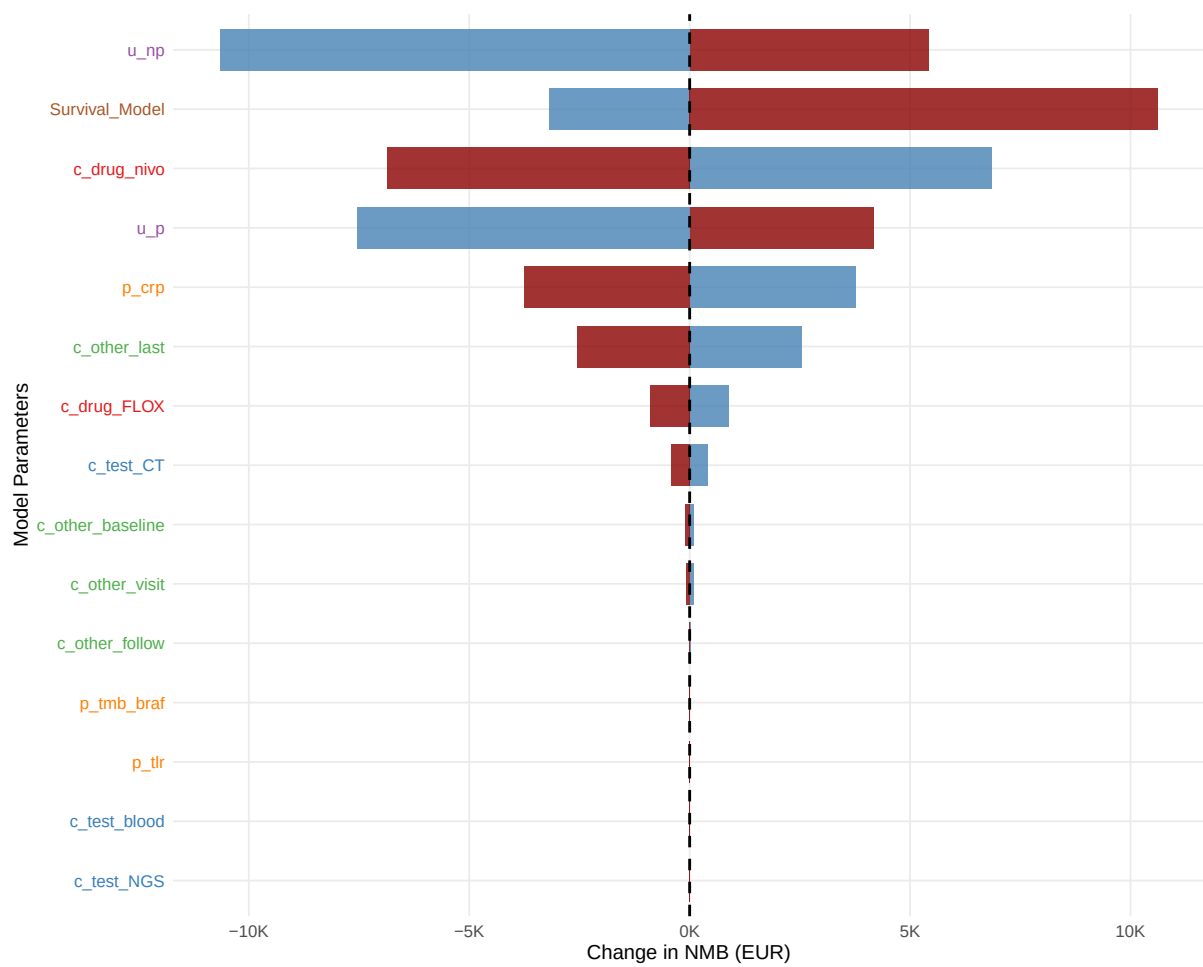


Figure 2: Tornado Diagram - CRP-guided Strategy One-Way Sensitivity Analysis

4.3 TLR-guided Strategy

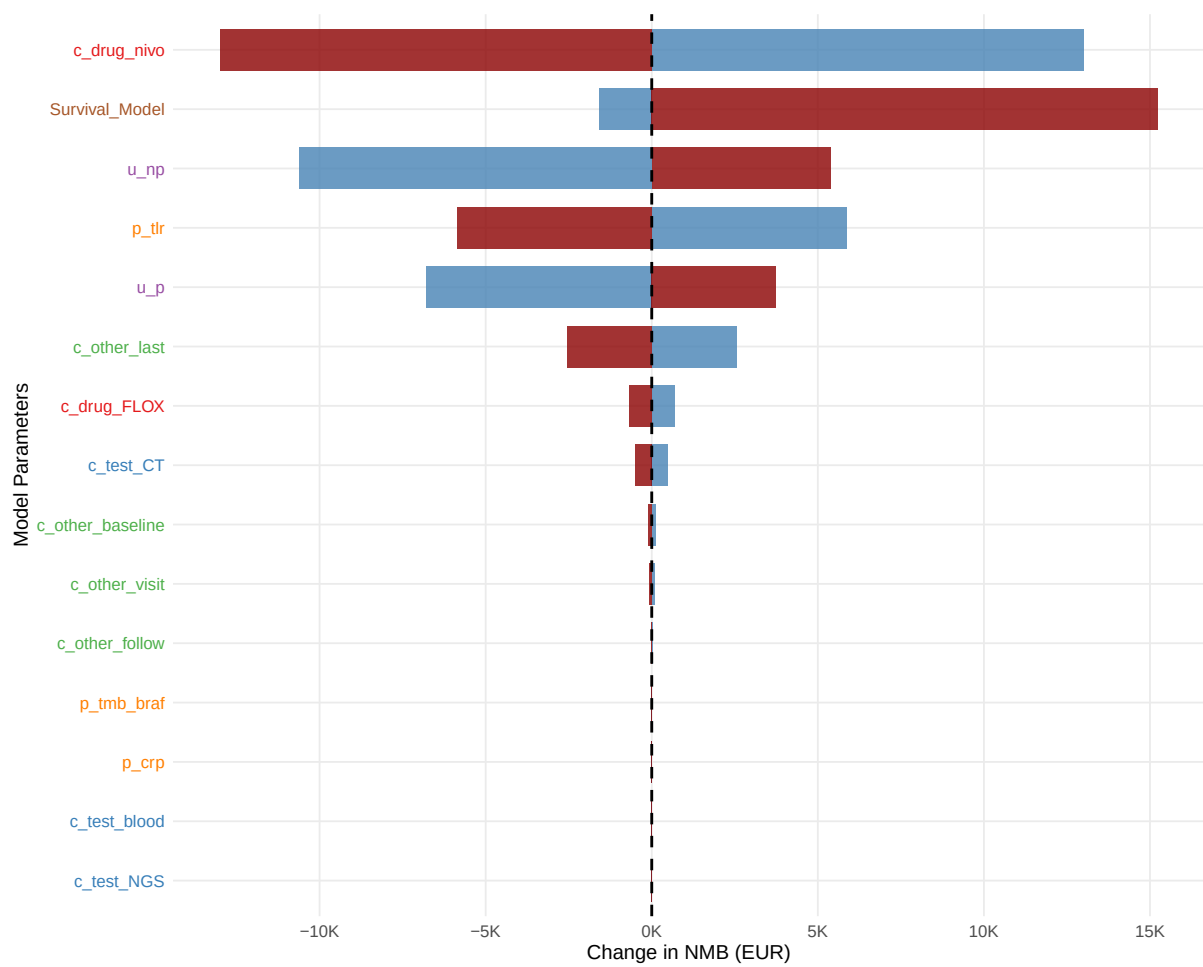


Figure 3: Tornado Diagram - TLR-guided Strategy One-Way Sensitivity Analysis

4.4 TMB/BRAF-guided Strategy

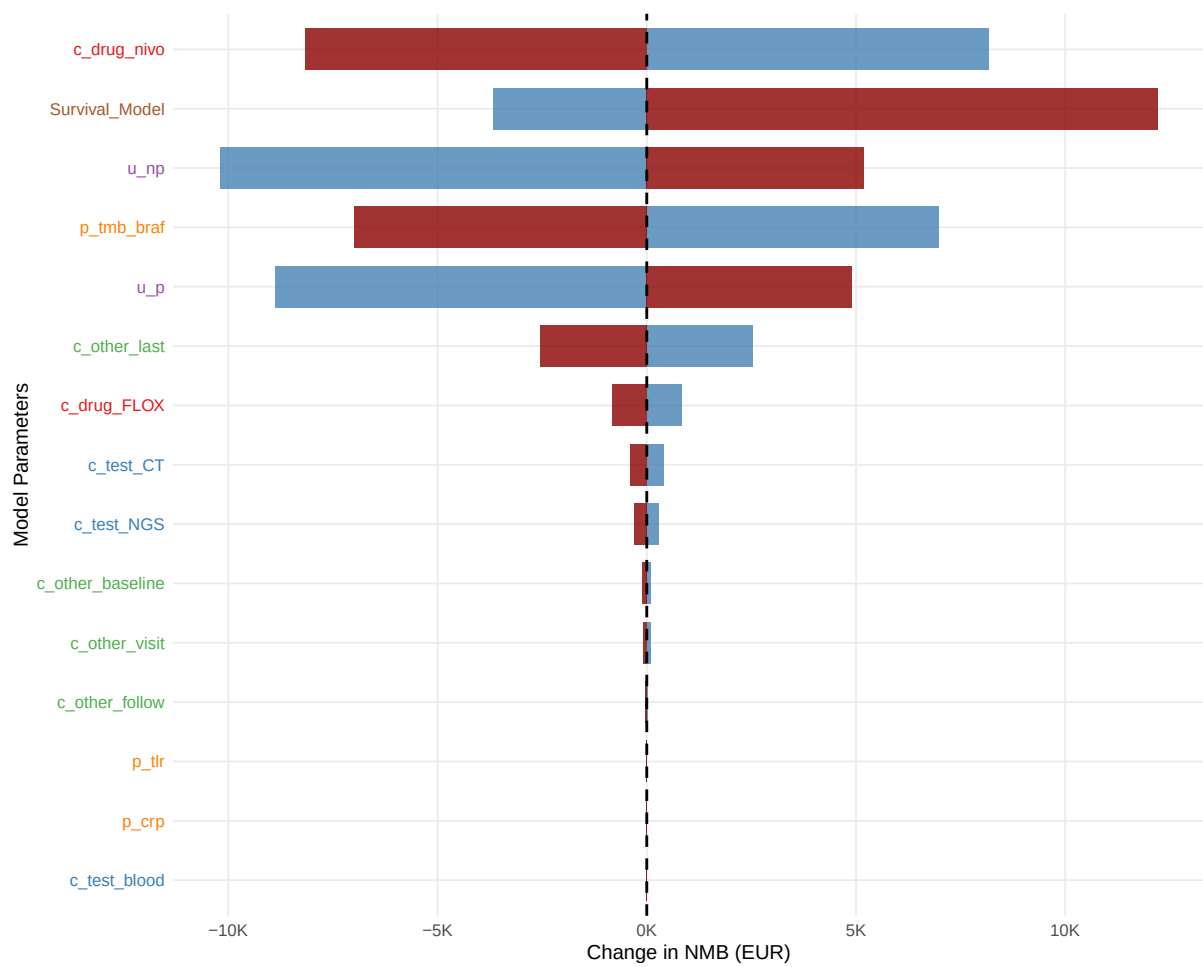


Figure 4: Tornado Diagram - TMB/BRAF-guided Strategy One-Way Sensitivity Analysis

5 Line Plot Sensitivity Analysis

5.1 Net Monetary Benefit

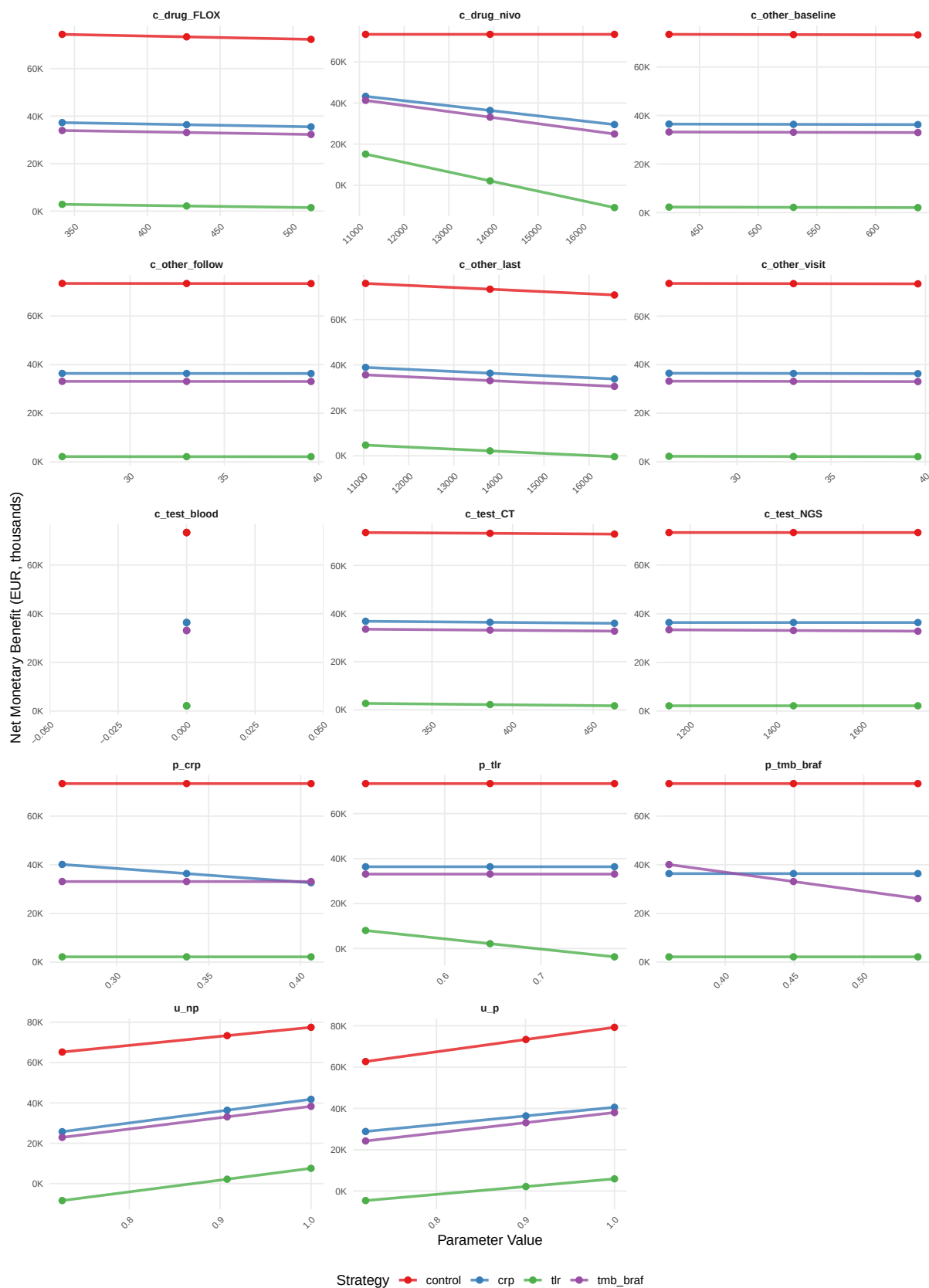


Figure 5: Parameter Sensitivity - Net Monetary Benefit by Parameter

5.2 Total Costs

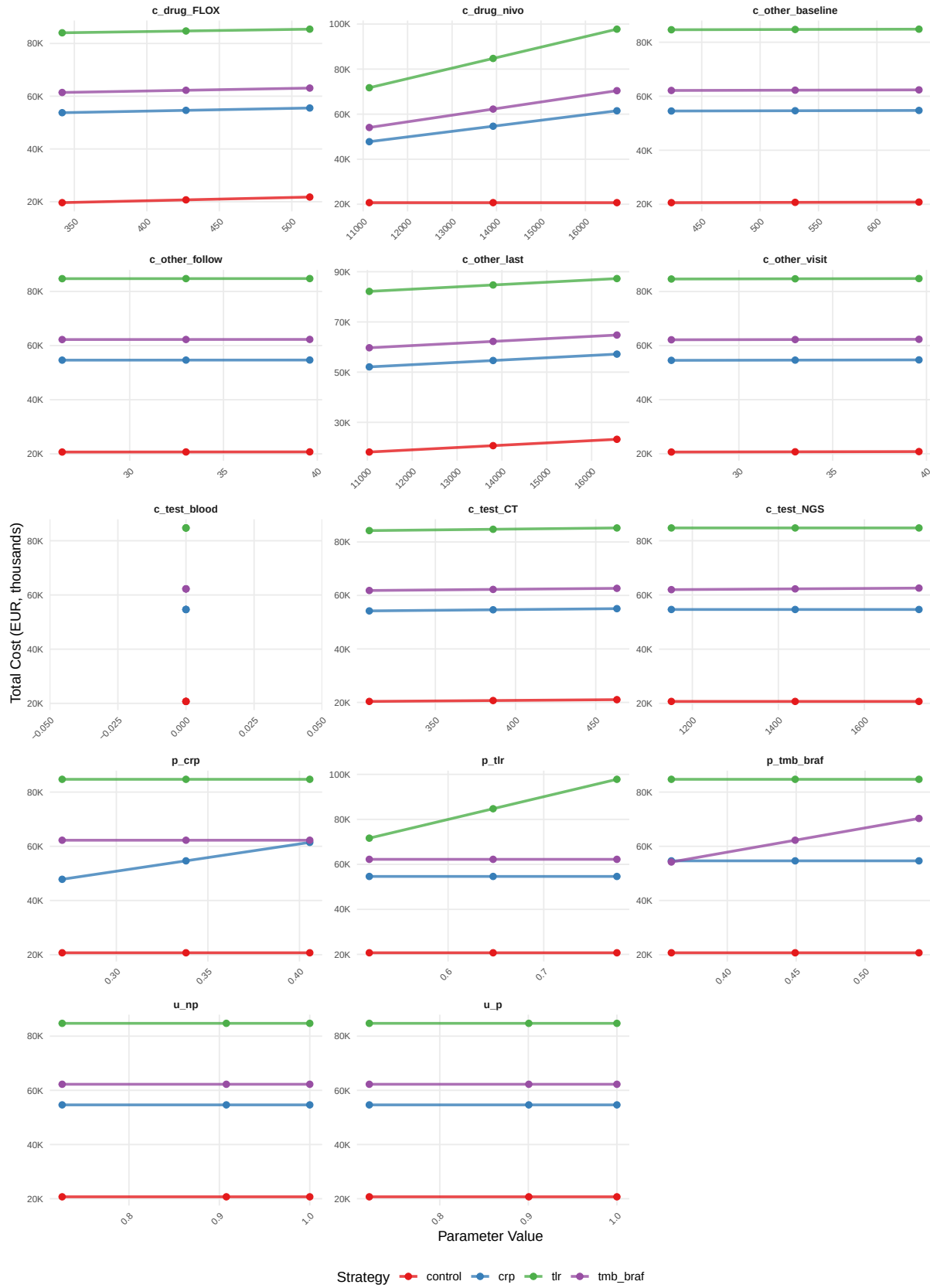


Figure 6: Parameter Sensitivity - Total Costs by Parameter

5.3 Total Effectiveness

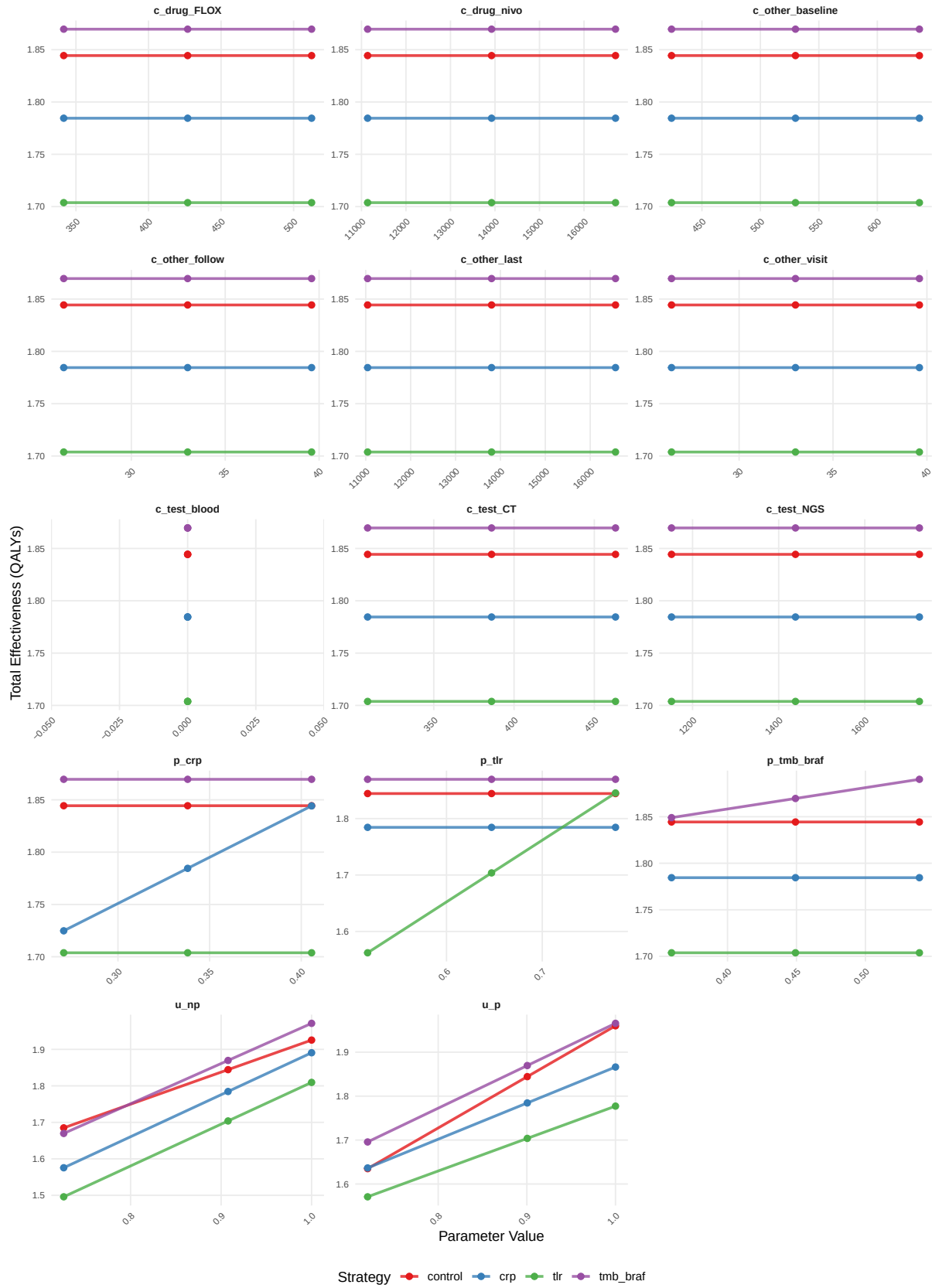


Figure 7: Parameter Sensitivity - Total Effectiveness (QALYs) by Parameter

6 Robustness Analysis

6.1 Strategy Switching Analysis

The optimal strategy (control) remains unchanged across all parameter variations tested. This indicates robust decision-making within the $\pm 20\%$ parameter variation range.

6.2 Parameter Impact Rankings

Table 2: Top 3 Most Influential Parameters by Strategy

Strategy	Rank	Parameter	NMB Range (EUR K)
Standard of Care	1	Utility (P)	16.6
	2	NA	13.1
	3	Utility (PF)	12.3
CRP-guided	1	Utility (PF)	16.1
	2	NA	13.8
	3	Nivolumab cost	13.7
TLR-guided	1	Nivolumab cost	26.0
	2	NA	16.8
	3	Utility (PF)	16.0
TMB/BRAF-guided	1	Nivolumab cost	16.3
	2	NA	15.9
	3	Utility (PF)	15.4

Note: Impact represents the absolute difference in Net Monetary Benefit between the minimum and maximum parameter values for each strategy.

7 Survival Model Structural Sensitivity

7.1 Overview

This section examines how the choice of parametric survival distribution affects cost-effectiveness results. The base case uses the best-fitting distribution selected by AIC. All 9 fitted distributions are tested with the constraint that OS and PFS use the same distribution to ensure methodological consistency.

Distributions tested:

1. Exponential
2. Weibull
3. Weibull PH (proportional hazards)
4. Log-logistic
5. Log-normal
6. Gamma (base case)
7. Gompertz
8. Generalized Gamma
9. Generalized F

7.2 NMB by Parametric Distribution

Table 3: Net Monetary Benefit by Parametric Distribution and Strategy

Distribution	control	crp	tlr	tmb_braf	control Rank	crp Rank	tlr Rank	tmb_braf Rank
exponential	84210	46993	17401	45317	1.0	1.0	1.0	1.0
weibull	77211	35422	765	31969	2.5	7.5	7.5	7.5
weibullph	77211	35422	765	31969	2.5	7.5	7.5	7.5
llogis	71116	40830	5583	36407	9.0	2.0	3.0	2.0
lognormal	71352	39237	7684	35374	8.0	3.0	2.0	3.0
gamma	73357	36373	2164	33104	5.0	4.0	4.0	6.0
gompertz	76923	33187	586	29426	4.0	9.0	9.0	9.0
gengamma	73216	36295	1871	33258	7.0	6.0	5.0	4.0
genf	73267	36298	1849	33255	6.0	5.0	6.0	5.0

7.3 Impact Summary

Table 4: Survival Model Impact on NMB by Strategy

Strategy	Base NMB	Min NMB	Max NMB	Range
Standard of Care	73357	71116 (llogis)	84210 (exponential)	13093
CRP-guided	36373	33187 (gompertz)	46993 (exponential)	13806
TLR-guided	2164	586 (gompertz)	17401 (exponential)	16815
TMB/BRAF-guided	33104	29426 (gompertz)	45317 (exponential)	15891

7.4 Interpretation

The survival model choice represents **structural uncertainty** in the cost-effectiveness analysis. Different parametric distributions make different assumptions about how hazard rates change over time:

- **Exponential:** Constant hazard (no change over time)
- **Weibull/Gompertz:** Monotonically increasing or decreasing hazard
- **Log-logistic/Log-normal:** Non-monotonic hazard (can increase then decrease)

- **Gamma/Generalized Gamma:** Flexible hazard shapes

The range of NMB across distributions indicates sensitivity to these extrapolation assumptions. A large range suggests that long-term survival projections are particularly influential and that the choice of parametric form matters for decision-making.

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